

Digital Image Processing

ASSIGNMENT -4

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Histogram Equalization

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

def histogram_equalization(gray_levels, nk):

    MN = np.sum(nk)
    pdf = nk / MN
    cdf = np.cumsum(pdf)
    L = len(gray_levels)
    Sk=(L - 1) * cdf
    Sk_roundOff = np.round(Sk).astype(int)

    new_histogram = np.zeros(L)
    for i in range(len(Sk)):
        new_histogram[Sk_roundOff[i]] += nk[i]

    df = pd.DataFrame({
        'GrayLevel': gray_levels,
        'nk': nk,
        'PDF (nk/MN)': pdf,
        'CDF': cdf,
        'Sk':np.round(Sk,2),
        'Sk (roundOff)': Sk_roundOff
```

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    })

    df_new=pd.DataFrame({
        'GrayLevel': gray_levels,
        'new nk':new_histogram
    })
    pd.set_option('display.float_format', lambda x: '%.6f' % x)
    return df.to_string(index=False,
justify='center'),df_new.to_string(index=False,
justify='center'), new_histogram

gray_levels = np.array([0, 1, 2, 3, 4, 5, 6, 7])
nk = np.array([790, 1023, 850, 656, 329, 245, 122, 81])

df_result,df_new_result, new_hist =
histogram_equalization(gray_levels, nk)

print(df_result)

fig, axs = plt.subplots(1, 2, figsize=(12, 5))

axs[0].bar(gray_levels, nk, color='red', alpha=0.7,
edgecolor='black')
axs[0].set_xlabel('GrayLevel')
axs[0].set_ylabel('Frequency (nk)')
axs[0].set_title('Initial Histogram')
axs[0].set_xticks(gray_levels)
axs[0].grid(axis='y', linestyle='--', alpha=0.7)

```

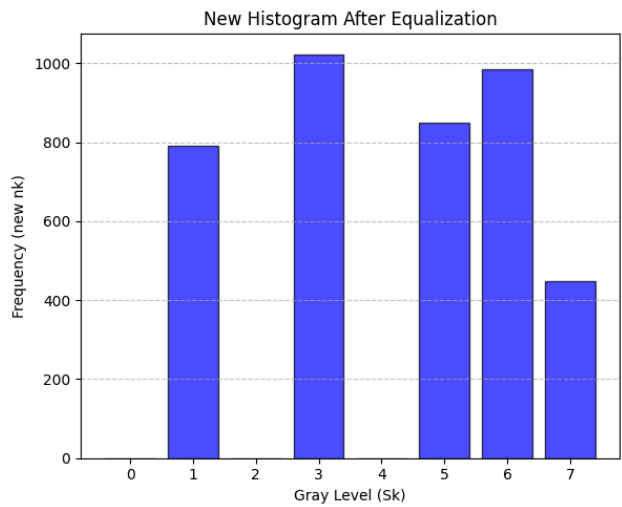
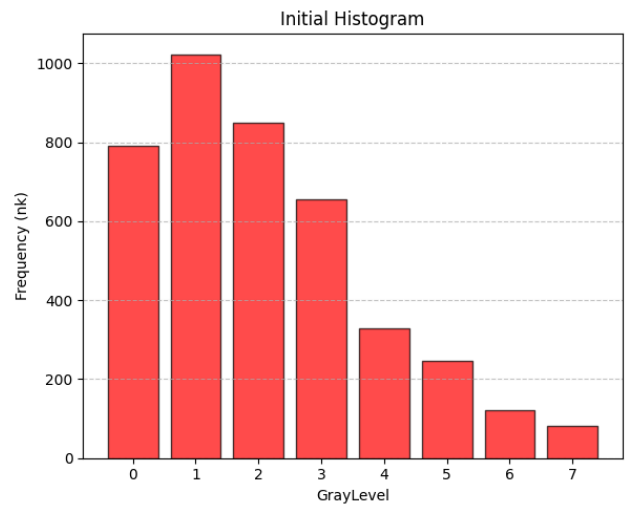
```

axs[1].bar(gray_levels, new_hist, color='blue', alpha=0.7,
edgecolor='black')
axs[1].set_xlabel('Gray Level (Sk)')
axs[1].set_ylabel('Frequency (new nk)')
axs[1].set_title('New Histogram After Equalization')
axs[1].set_xticks(gray_levels)
axs[1].grid(axis='y', linestyle='--', alpha=0.7)

plt.tight_layout()
plt.show()
print("New Table")
print(df_new_result)

```

GrayLevel	nk	PDF (nk/MN)	CDF	Sk	Sk (roundOff)
0	790	0.192871	0.192871	1.350000	1
1	1023	0.249756	0.442627	3.100000	3
2	850	0.207520	0.650146	4.550000	5
3	656	0.160156	0.810303	5.670000	6
4	329	0.080322	0.890625	6.230000	6
5	245	0.059814	0.950439	6.650000	7
6	122	0.029785	0.980225	6.860000	7
7	81	0.019775	1.000000	7.000000	7



New Table

GrayLevel	new nk
0	0.000000
1	790.000000
2	0.000000
3	1023.000000
4	0.000000
5	850.000000
6	985.000000
7	448.000000